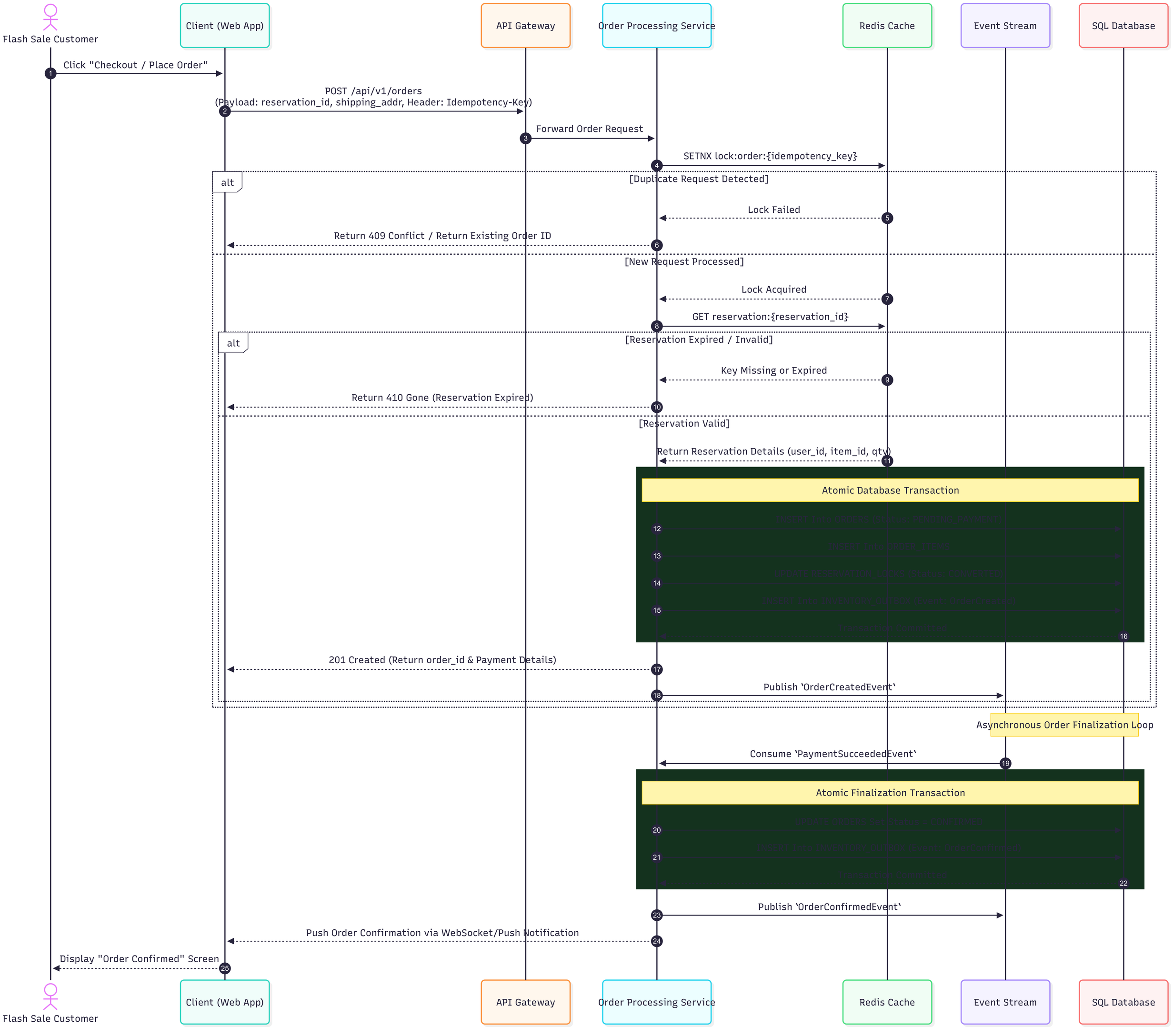

Flash Sale Customer
Client (Web App)
API Gateway
Order Processing Service
Redis Cache
Event Stream
SQL Database
1 Click "Checkout / Place Order"
2 POST /api/v1/orders
(Payload: reservation_id, shipping_addr, Header: Idempotency-Key)
3 Forward Order Request
4 SETNX lock:order:{idempotency_key}
alt
[Duplicate Request Detected]
5 Lock Failed
6 Return 409 Conflict / Return Existing Order ID
[New Request Processed]
7 Lock Acquired
8 GET reservation:{reservation_id}
alt
[Reservation Expired / Invalid]
9 Key Missing or Expired
10 Return 410 Gone (Reservation Expired)
[Reservation Valid]
11 Return Reservation Details (user_id, item_id, qty)
Atomic Database Transaction
12 INSERT into ORDERS (Status: PENDING_PAYMENT)
13 INSERT into ORDER_ITEMS
14 UPDATE RESERVATION_LOCKS (Status: CONVERTED)
15 INSERT into INVENTORY_OUTBOX (Event: OrderCreated)
Transaction Committed
16
17 201 Created (Return order_id & Payment Details)
18 Publish 'OrderCreatedEvent'
Asynchronous Order Finalization Loop
19 Consume 'PaymentSucceededEvent'
Atomic Finalization Transaction
20 UPDATE ORDERS Set Status = CONFIRMED
21 INSERT into INVENTORY_OUTBOX (Event: OrderConfirmed)
Transaction Committed
22
23 Publish 'OrderConfirmedEvent'
24 Push Order Confirmation via WebSocket/Push Notification
25 Display "Order Confirmed" Screen
Flash Sale Customer
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